SAIDI and SAIFI Benchmark Comparison

Purpose. This note provides a transparent comparison of SAIDI and SAIFI provisions across the 12 most populous Indian states and selected international reference systems. For Indian states, the table distinguishes between explicit regulatory standards/targets, reporting requirements, and cases where no verified numeric benchmark has yet been identified. For international comparators, the figures shown are observed system performance rather than regulatory standards.

Indian states covered. Uttar Pradesh, Bihar, Maharashtra, West Bengal, Madhya Pradesh, Rajasthan, Tamil Nadu, Gujarat, Karnataka, Andhra Pradesh, Odisha and Telangana.

Master comparison table

Geography	Status of figure	SAIDI - original value	SAIFI - original value	Scope / coverage	Key methodological qualifications	Reference / source
India	Range of available state-level regulatory benchmarks	600–3,600 min/customer/year	24–144 interruptions/customer/year	Range derived from available numerical regulatory standards/targets identified for Andhra Pradesh, Madhya Pradesh and Rajasthan; not a national regulatory standard	Represents the minimum and maximum among identified state-level numerical benchmarks, not national minimum/ maximum performance. States without a verified numerical benchmark are excluded from the range.	Listed below
United States	Actual observed performance	131.6 min/customer/year	1.065 times/year	U.S. distribution utilities; IEEE methodology	Figure shown excludes Major Event Days. EIA separately publishes all-events figures, which are materially higher.	EIA Electric Power Annual, 2024
Japan	Actual observed performance	24 min/customer/year	0.13 interruptions/customer/year	Nationwide low-voltage customers	Reported total includes forced and planned interruption components. FY2024: SAIFI 0.11 forced + 0.02 planned; SAIDI 21 min forced + 3 min planned.	OCCTO, FY2024

Germany	Actual observed performance	11.7 min/customer/ year	Not reported in selected source	Electricity end customers covered by Bundesnetzagentur interruption reporting	Planned interruptions and force majeure excluded; only interruptions >3 min counted.	Bundesnetzagentur, 2024
Singapore	Actual observed performance	0.32 min/customer/ year	0.022 interruptions/customer/year	National electricity system	Compact city-state; useful as a reference point but structurally very different from an Indian state.	EMA, 2025
Brazil	Actual observed performance	618 min/customer/ year	4.9 interruptions/customer/year	Distribution system	Observed national distribution performance, not a regulatory standard.	PSR, 2024

Methodological notes

Annualisation. Where an Indian regulation specifies a quarterly, half-yearly or other non-annual standard, the annual value is a simple arithmetic conversion provided only to aid visual comparison. It does not represent the legally applicable compliance period.

Comparability. These figures are not strictly like-for-like. Indian entries principally describe regulatory standards, targets or reporting requirements, whereas the international entries describe observed performance. Jurisdictions also differ in their treatment of planned interruptions, major-event days / force majeure, minimum interruption duration, voltage levels, agricultural feeders and consumer coverage. These qualifications should remain visible alongside the numbers rather than being normalised away.

Interpretation. A blank or 'no verified numeric benchmark' entry is intentional. It indicates that the current first-pass review did not identify a defensible operative numeric standard in the core regulation/order reviewed. It should not be read as evidence that the regulator does not monitor SAIDI/SAIFI.

References for India (state-wise)

1. **Uttar Pradesh — UPERC:** current applicable regulatory framework; your table does not yet contain a specific document link. No numerical benchmark was verified in this pass. Pasted markdown
2. **Bihar — BERC Standards of Performance Regulations, as amended:** BERC SoP Regulations Pasted markdown

3. **Maharashtra — MERC Electricity Supply Code and Standards of Performance Regulations, 2021:** MERC Supply Code & SoP Regulations 2021 Pasted markdown
4. **West Bengal — WBERC:** WBERC official website. Order-level verification remains pending. Pasted markdown
5. **Madhya Pradesh — MPERC FY2026-27 target:** MPERC State DISCOMs ARR and Retail Supply Tariff Order. This provides the trajectory culminating in **60 h SAIDI / 90 SAIFI** for cities ≥1 lakh population. Pasted markdown
6. **Rajasthan — RERC Standards of Performance framework:** RERC SoP framework. Your table records **30 h/6 months SAIDI and 20 interruptions/6 months SAIFI**, annualised to 3,600 minutes and 40 interruptions. Pasted markdown
7. **Tamil Nadu — TNERC Distribution Standards of Performance Regulations:** TNERC Distribution SoP Regulations Pasted markdown
8. **Gujarat — GERC Standards of Performance Regulations, 2023:** GERC SoP Regulations 2023 Pasted markdown
9. **Karnataka — KERC:** current framework/order-level verification remains pending; your table does not currently contain a specific source URL. Pasted markdown
10. **Andhra Pradesh — APERC Practice Directions / Standards of Performance framework:** APERC Practice Directions / SoP framework. The category-specific values are **600/1,200/2,400 minutes SAIDI** and **24/72/144 SAIFI** for EHT, 33 kV industrial and 11 kV industrial categories respectively. Pasted markdown
11. **Odisha — OERC Standards of Performance framework:** OERC SoP framework Pasted markdown
12. **Telangana — TGERC/TSERC Standards of Performance framework:** TGERC/TSERC SoP framework